File Handling in Java Demonstration

Title: File Handling – Reading and Writing Files in Java and Finding the Frequency of Patterns

1. Objective / Aim

To write the given text into a file named "sample.txt", read the file, and find the frequency of the patterns "pe" and "pi".

2. System Requirements

Hardware: Standard PC/Laptop, Intel Core i3/i5 processor or equivalent, minimum 4 GB RAM, 500 MB free storage, keyboard and monitor.

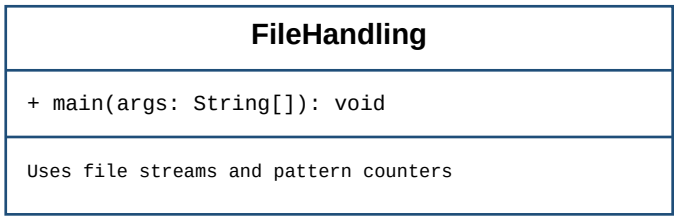
Software: Windows/Linux/macOS, JDK 17 or JDK 21, Visual Studio Code/Eclipse or any Java IDE, Command Prompt/Terminal.

3. Algorithm / Logic Description

1. Start the program.
2. Create/open sample.txt.
3. Write the given text using FileWriter and close the writer.
4. Read the file using FileReader and BufferedReader.
5. Combine the lines and convert the content to lowercase.
6. Count all adjacent occurrences of "pe" and "pi".
7. Display both frequencies.
8. Stop the program.

4. UML Class Diagram

FileHandling class and its main method:



5. Source Code

```
import java.io.*;

public class FileHandling {
    public static void main(String[] args) {
        String text = "Peter Piper picked a peck of pickled peppers\n"
                + "A peck of pickled peppers Peter Piper picked\n"
                + "If Peter Piper picked a peck of pickled peppers\n"
                + "Where's the peck of pickled peppers Peter Piper picked?";

        try {
            FileWriter fw = new FileWriter("sample.txt");
            fw.write(text);
            fw.close();

            BufferedReader br = new BufferedReader(
                    new FileReader("sample.txt"));

            StringBuilder content = new StringBuilder();
            String line;

            while ((line = br.readLine()) != null) {
                content.append(line).append("\\n");
            }
            br.close();

            String data = content.toString().toLowerCase();
            int peCount = 0;
            int piCount = 0;

            for (int i = 0; i < data.length() - 1; i++) {
                String pattern = data.substring(i, i + 2);
                if (pattern.equals("pe")) peCount++;
                if (pattern.equals("pi")) piCount++;
            }

            System.out.println("'pe' - no of occurrences - " + peCount);
            System.out.println("'pi' - no of occurrences - " + piCount);

        } catch (IOException e) {
```

```
            System.out.println("An error occurred: " + e.getMessage());
        }
    }
}
```

6. Compilation and Execution Commands

```
javac FileHandling.java
java FileHandling
```

7. Console Output Screenshot

Expected output from a successful run:

```
PS C:\Java-Projects> javac FileHandling.java
PS C:\Java-Projects> java FileHandling

'pe' - no of occurrences - 20

'pi' - no of occurrences - 12

PS C:\Java-Projects>
```

8. Explanation

• **FileWriter:** Writes the supplied text to sample.txt.

• **FileReader / BufferedReader:** Read the file sequentially, one line at a time.

• **StringBuilder:** Combines the lines read from the file.

• **toLowerCase():** Makes the pattern search case-insensitive.

• **substring(i, i + 2):** Extracts each adjacent pair for comparison.

• **IOException:** Handles file-related errors.